

HONGBO LI

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EDUCATION

Michigan State University

East Lansing, MI

Aug 2025 – Present

Doctoral Program in Statistics, College of Natural Science

- Selected PhD Courses: Linear Model Methodology, Theory of Probability I, Theory of Probability II, Mixed Models, Statistical Inference I

Johns Hopkins University

Baltimore, MD

Aug 2023 – May 2025

Master of Science in Financial Mathematics, Whiting School of Engineering

Major GPA in Math: 3.96/4.0

- Selected Courses: Nonlinear optimization (PhD), Advanced Financial Theory (PhD), Applied Statistics & Data Analysis, Stochastic Process, Monte Carlo, Machine Learning in Finance, Time Series, Bayesian Statistics

Huazhong University of Science and Technology

Wuhan, China

Sep 2018 – Jun

2022

Bachelor of Science in Statistics, School of Mathematics and Statistics

Major GPA in Math and CS: 3.8/4.0

- Selected Courses: Real Analysis, Probability Theory, Modern Algebra, Mathematical Statistics, Multivariate Statistics, Operations Research, Optimization Method, Partial Differential Equations, Advanced Numerical Methods
- Thesis: “Robust Facial Recognition based on CNN”

RESEARCH INTERESTS

Stochastic approximation, Computational algorithms in optimization, Monte Carlo methods, Reinforcement learning, Statistical machine learning

PUBLICATIONS AND MANUSCRIPTS

[1] **Hongbo Li** and James C. Spall.

Zeroth-Order Langevin Monte Carlo via SPSA under Noisy Function Measurements.

Preprint, arXiv:2608.07837, 2026. <https://arxiv.org/abs/2608.07837>. Journal manuscript in final preparation.

[2] **Hongbo Li** and James C. Spall.

Zeroth-Order Langevin Sampling for Bayesian Applications.

Accepted at the 4th IEEE International Conference on Artificial Intelligence, Blockchain and Internet of Things (AIBThings), 2026.

[3] Meiqi Liu, Zhuoqun Huang, **Hongbo Li**, Young-geun Kim, and Yue Xing.

How to Enhance Downstream Adversarial Robustness (Almost) Without Touching the Pre-trained Foundation Model?

Manuscript in preparation.

RESEARCH EXPERIENCE

Zeroth-Order Langevin Monte Carlo via SPSA under Noisy Function Measurements

Jan 2025 – Present

Supervised by Prof. James Spall, Johns Hopkins University

- Motivation & method:** Bridged **gradient-based** samplers and **gradient-inaccessible** settings in practice (simulation-based inference, model-free reinforcement learning, and stochastic control) by proposing Langevin Monte Carlo-

Simultaneous Perturbation Stochastic Approximation (**LMC-SPSA**) in noisy settings—gradient-free Langevin sampler with only **two** noisy function queries per iteration (**dimension-independent**).

Theoretical development:

- **Convergence:** Proved Wasserstein-2 convergence of LMC-SPSA under noise and resolved **open step size question** by deriving **diminishing-step size window** ensuring convergence, extending prior constant-step size analysis.
- **Sharper dimension dependence:** Improved theoretical dominant dimension dependence of the Wasserstein-2 error bound **from quartic to quadratic**, with numerical results supporting the theoretical scaling.
- **Sharper oracle complexity:** Reduced **theoretical noisy-oracle complexity** needed to reach target Wasserstein accuracy relative to Roy et al.'s zeroth-order LMC method (Bernoulli, 2022), improving accuracy dependence **from fourth order to second order**.
- **Theory-guided shrinking schedule:** Derived **coupled diminishing** step-size and perturbation schedules that balance Langevin discretization error, SPSA bias, and residual error terms, giving practical tuning rules for LMC-SPSA.
- **Budget-matched numerical validation:** Conducted oracle-call-matched experiments against LMC-Finite Difference Stochastic Approximation and Roy et al.'s zeroth-order LMC, showing that LMC-SPSA attains lower and more stable MSE and higher-order moment errors under fixed function-evaluation budgets.

Bayesian applications and empirical evaluation:

- **Black-box Bayesian sampling:** Adapted LMC-SPSA to black-box Bayesian posterior sampling settings where gradients are unavailable and only posterior-density evaluations are accessible.
- **Empirical comparison:** Designed fixed-budget experiments on Bayesian classification on MNIST 3-vs-8 and Bayesian regression on UCI Energy Efficiency data set, showing that LMC-SPSA outperforms random-walk Metropolis-Hastings and diagonal adaptive Metropolis under matched posterior-density evaluation budgets.

Downstream Adversarial Robustness of Frozen Foundation Models

Jan 2026 – May 2026

Supervised by Prof. Yue Xing and Prof. Young-geun Kim, Michigan State University

- **Motivation & method:** Investigated whether label-free robust **preprocessing** and **generated training data** can improve downstream adversarial robustness in frozen foundation-model pipelines **without full model fine-tuning**.
- **Synthetic-data evaluation pipeline:** Extended CIFAR-10 pipelines to support DDPM/DDIM synthetic data and synthetic-real mixtures across preprocessor training, linear-head training, and PGD-based robustness evaluation.
- **Synthetic data quality analysis:** Designed CIFAR-10 ablation studies with DDPM/DDIM-generated datasets of varying quality, showing that higher complementarity, coverage, and Inception Score, together with lower FID, are positively associated with downstream robust accuracy, and identified an intermediate synthetic-real mixing ratio that outperforms fully real or fully synthetic data.
- **Wasserstein proof refinement:** Refined the downstream adversarial-loss proof by reducing the conditional Wasserstein term to clean-adversarial encoder displacement under deterministic-Dirac representations, and derived a lower-bound interpretation via optimal transport between aggregated latent distributions.

Additional Research Project

Fine-tuning FinBERT with Few-Shot Weak Supervision for Financial News Sentiment

Aug 2024 – Feb 2025

Supervised by Prof. Helyette Geman, Johns Hopkins University

- **Motivation & method:** Applied **few-shot in-context labeling** with gold examples to generate **weak labels** at scale, using consistency voting and confidence filtering for **denoising**; then fine-tuned FinBERT (LoRA) with temporal splits and probability calibration. The fine-tuned model outperformed the baseline FinBERT in accuracy.
- **Multi-ticker disambiguation:** Designed three datasets for multi-ticker articles and found that **most-mentioned** ticker rule gave the highest sentiment-classification accuracy and strongest 2-day Granger-prediction.

PRESENTATION

IEEE AIBThings 2026, Central Michigan University	Mount Pleasant, MI	Sep 2026
Accepted presentation: Zeroth-Order Langevin Sampling for Bayesian Applications		
LAMP 2026 Workshop, Toyota Technological Institute at Chicago	Chicago, IL	Aug 2026
Accepted presentation: Two Evaluations per Step: LMC-SPSA for Black-Box Bayesian Sampling		
Statistics and Data Science Workshop, Universidad de los Andes	Bogota, Colombia	Dec 2025
Accepted presentation: Zeroth-Order Langevin Monte Carlo via SPSA under Noisy Function Measurements		
Financial Mathematics Seminar, Johns Hopkins University	Baltimore, MD	Oct 2024
Presentation for summer intern as Quantitative Researcher		

TEACHING EXPERIENCE

Teaching Assistant	Michigan State University	Aug 2025 – Dec 2025
Taught three weekly sections of STT 200 (Statistical Methods)		

INDUSTRY EXPERIENCE

China Merchants Securities Co., Ltd.	Beijing, China	Jul 2024 – Nov 2024
Quantitative Researcher Summer Intern		

- Applied **Kalman filter** to denoise intraday signals and enhance accuracy of trading decisions, improved win rate, profit-loss ratio, and full-period information ratio vs. out-of-sample baseline
- Built **cointegration-based** statistical arbitrage framework: screened economically linked FX pairs, tested cointegration (Engle-Granger, Johansen), estimated hedge ratios and mean-reversion half-life, then validated out-of-sample with rolling retests

SCHOLARSHIPS & AWARDS

- Dean's Scholarship, Huazhong University of Science and Technology
2018
- National Innovation Training Program for Undergraduate Research
2022

COMPUTER SKILLS

- Programming Skills: Python (NumPy, Pandas, Scikit-learn, PyTorch), MATLAB, R, C++, MySQL
- Machine Learning: Transformers, Diffusion models, Adversarial training, Bayesian computation